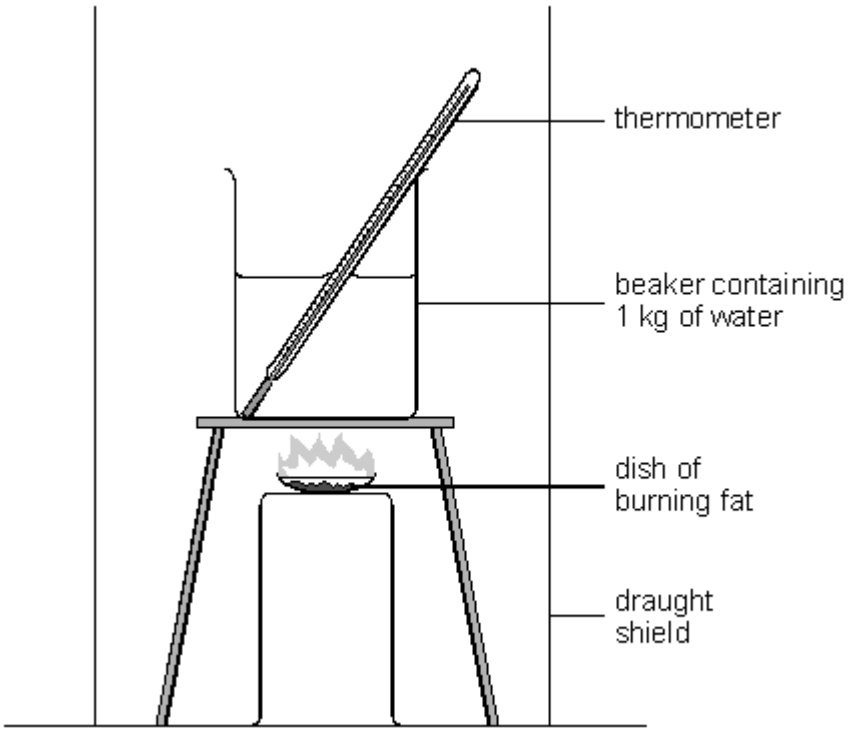
Q1. (a) Mammals use fat as a long-term store of energy.

(i) The diagram shows some simple apparatus for investigating the amount of energy released by burning fat.



4.2 kJ of energy will raise the temperature of 1 kg of water by 1°C.
1 g of fat contains 38.5 kJ of energy.

Calculate the rise in temperature of 1 kg of water if 0.5 g of fat is burned.
Show your working.

……

……

……

……

2 marks

(ii) Fat is stored in layers beneath the skin. This fat is part of the body's energy reserve. Give another function of the layers of fat.

……

……

1 mark

(b) Mammals use glucose as a fuel in respiration in their cells. The word equation for respiration is:

glucose + oxygen → carbon dioxide + water

Mammals with a high rate of respiration need to have a high heart rate. Explain why.

……

……

……

……

2 marks
Maximum 5 marks

Mark schemes

Q1.

(a) (i) energy in 0.5 g of fat = 19.25 kJ

*accept '38.5 kJ **or** 1 g of fat gives a rise in temperature of 9.2°C' unit not required for the mark*

1

4.6°C

accept answers in the range 4.56 – 4.60°C

accept the correct answer with no working for both marks

the unit is required for the mark

1

(ii) any **one** from

- it is an insulator
- it reduces heat loss

accept 'to keep the heat in'
***or** 'to keep the body temperature constant'*

accept 'it protects the organs'

*do **not** accept 'it stops heat loss' **or** 'it keeps us warm'*

1

(b) any **two** from

- to transport oxygen quickly

accept 'to transport more oxygen'
***or** ' oxygen is used up quickly' **or** 'more oxygen is needed'*

- to remove carbon dioxide quickly

accept 'carbon dioxide is produced quickly'
***or** 'to transport more carbon dioxide'*
***or** 'to get rid of waste more quickly'*

- to transport glucose quickly

*accept 'glucose is used up quickly' **or** 'to transport more glucose'*

accept 'to remove water quickly'
accept 'they have small hearts'

2

[5]